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GEOPHYSICS

The East European Craton in the Mesoproterozoic: New Key Paleomagnetic Poles

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At the present time, the "key poles" satisfying several criteria [1] are used in reconstructions of the Precambrian supercontinents: paleomagnetic measurements have to be made on magmatic rocks (mainly, dykes and sills) of age, obtained by applying the U–Pb method to baddeleite with an accuracy of no less ±20 Ma; the thermoremanent nature of magnetization, whose originality is confirmed by tests of paleo-

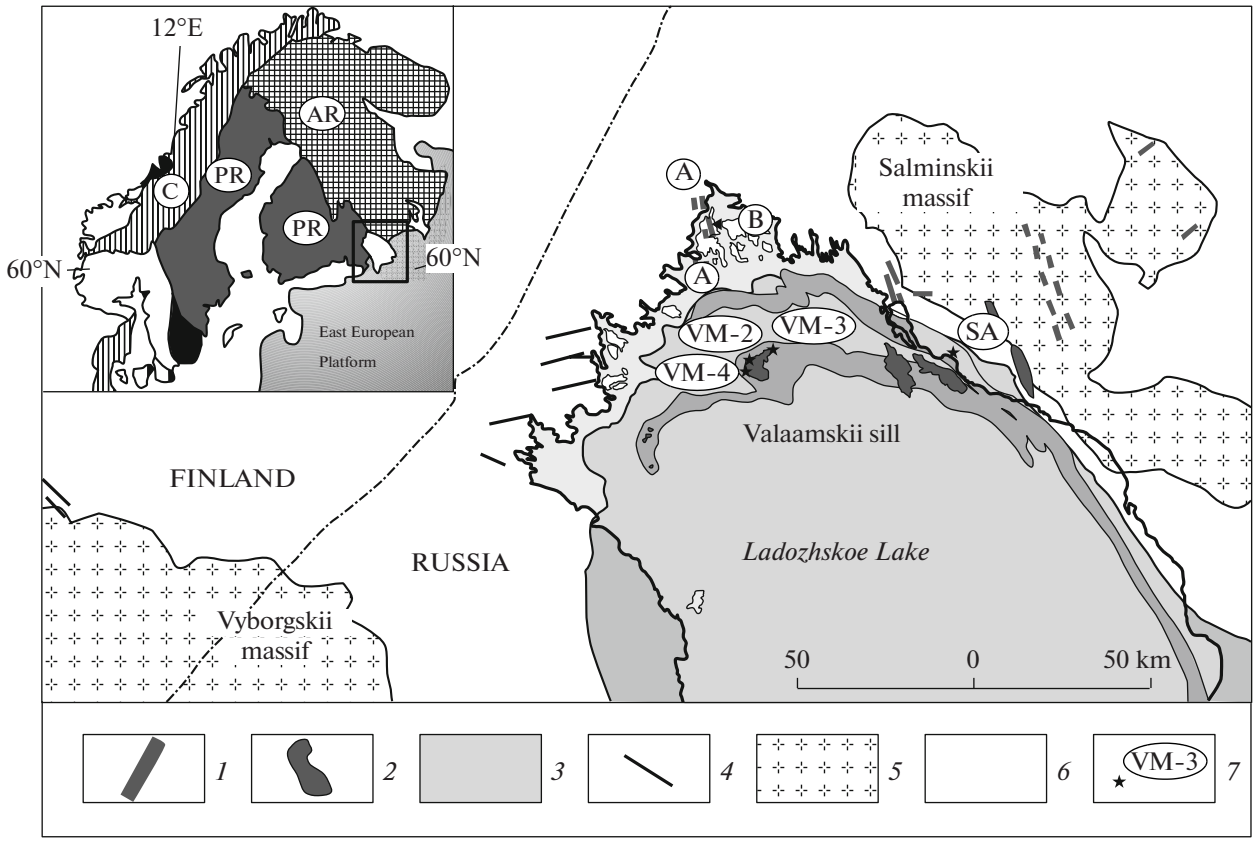


Fig. 1. Schematic geological map of North Priladozhie. In the display box are the main tectonic zones of the Fenno-Scandinavia shield: AR–Achaean Fenno-Carelian granite-gneiss greenstone protocontinent; PR–Paleoproterozoic Svecosfennian orogen; C–Caledonides. *1*—Mesoproterozoic dykes; *2*—Mesoproterozoic (Valaamsky) sill; *3*—Volcanogenic-terrigenous rocks of the Salminskaya suite; *4*—Dolerite dykes, associated with massives of rapakivi granites; *5*—Massives of rapakivi granites (1.62–1.54 Ma); *6*—Achaean and Proterozoic crust; *7*—Localities of sampling.

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Table 1. Paleomagnetic directions of the high-temperature component of magnetism of the rock studied

Object	Polarity	Geographic reference system				
		B/N	*Dec*	*Inc*	*K*	α_{95}
NORTH PRILADOZHIE						
Salma basalts (SA)	*N*	3	47.5	−24.1	43.1	19.0
	R	22	212.5	13.6	20.1	7.1
Average direction	*N* + *R*	**1/25**	**214.1**	**15.5**	**19.5**	**6.7**
Valaamskii sill (VA)	*N*	**3/49**	**29.9**	**−10.8**	**56.1**	**2.7**
Dykes of the *A*-type (A)	*N*	**4/166**	**38.2**	**−20.0**	**142.1**	**6.4**
Dykes of the *B*-type (B)	*N*	33	30.9	−18.6	230.0	6.1
	R	5	202.8	29.7	25.2	15.5
Average direction	*N* + *R*	**4/38**	**28.3**	**−20.5**	**504.2**	**5.5**
NORTH PRILADOZHIE		**4/278**	**32.5**	**−16.7**	**168.6**	**7.1**
SOUTHERN URALS						
Radostnyi Quarry (RD)	*N*	20	61.3	−40.7	22.2	7.1
	R	3	246.0	22.3	13.9	34.5
Average direction	*N* + *R*	**23**	**62.1**	**−38.1**	**19.4**	**7.1**
Mount Kapkatash (KP)	*N*	8	77.7	−17.2	15.2	14.7
	R	4	230.2	18.6	1.6	23.9
Average direction	*N* + *R*	12	71.0	−18.1	13.3	12.3
Average direction in stratigraphic reference system		**12**	**49.0**	**−42.2**	**13.3**	**12.3**
Bolshoi Klyuch Stream **(BK)**	*N*	**9**	**47.2**	**−21.6**	**61.7**	**6.6**
Berdyaush Quarry (BD)	*R*	11	225.7	−17.9	82.4	5.1
Berdyaush Quarry (stratigraphic reference system)	*R*	**11**	**223.9**	**51.9**	**82.6**	**5.1**
SOUTHERN URALS		4	56.3	−15.6	9.6	31.2
SOUTHERN URALS (stratigraphic reference system)		**4**	**50.8**	**−38.8**	**33.5**	**16.1**

Note: *B/N*—number of places of sampling/number of samples; Dec—declination; Inc—inclination; *K*—density; α_{95}—radius of confidence ellipse at 95% probability for the average direction. Average directions for every object are shown in bold type; key paleomagnetic poles used for data interpretation are shown in underlined bold type.

magnetic reliability (contact and inversion), has to be proved; averaging of secular variations, if the age period of the rocks sampled exceeds 40 000 years; and paleomagnetic poles, obtained for coeval rocks, not subjected to the late secondary alterations have to coincide among themselves in different parts of the same block.

In the Global Paleomagnetic Database for the East European Craton, there is a set of key poles in the intervals of 2.40–1.77 and 1.25–0.98 Ga, where as in the Mesoproterozoic rocks such poles were not recorded [1–3].

Recently, several new paleomagnetic determinations have been obtained for the Mesoproterozoic rocks in the East European Craton based on reliably dated objects [4, 5]. These poles, however, cannot be considered as the key ones, so they have been only obtained from one object (Valaamsky sill [4] and Salma basalts [5]), and there are no tests of their paleomagnetic reliability.

In order to fill in this gap, study of the Mesoproterozoic magmatic complexes in the northwestern part of the East-European Craton in North Priladozhie (South Carelia, Fig. 1) and the Bashkirskii anticlinorium (Southern Urals, Fig. 2) has been carried out.

In North Priladozhie the magmatic complexes are represented by gabbro-dolerites and monzonites of the Valaamskii sill, dykes of two generations (aphanitic dolerites of the *A*-type and high-ferriferous dolerites-sortavalites of the *B*-type), as well as by basalt flows of the Salminskaya suite (Fig. 1). The age of rocks of the Valaamskii sill was determined as 1458 ± 4 Ma [6], the age of dykes of sortavalites, 1452 ± 12 Ma [8]; and the age of Salma basalts, 1499 ± 68 Ma [7]. Moreover, in order to test the contact zone, the country granite-gneisses of the Ladozhskaya series have been sampled.

In the Southern Urals, the volcanites of the Bakalskaya and Mashakskaya suites of the Middle Riphean of the Bashkirskii Anticlinorium (Bolshoi Klyuch Stream and Mount Kapkatash) have been studied (Fig. 2.). Moreover, the basalt flows and interlaying tuffaceous sands have been sampled (Fig. 2). In addition, gabbro-dolerite dykes, intruding metamorphic rocks of the Taratash complex, and rapakivi granites have been studied in the Radostny and Berdyaush quarries, respectively. The age of the Mashakskaya

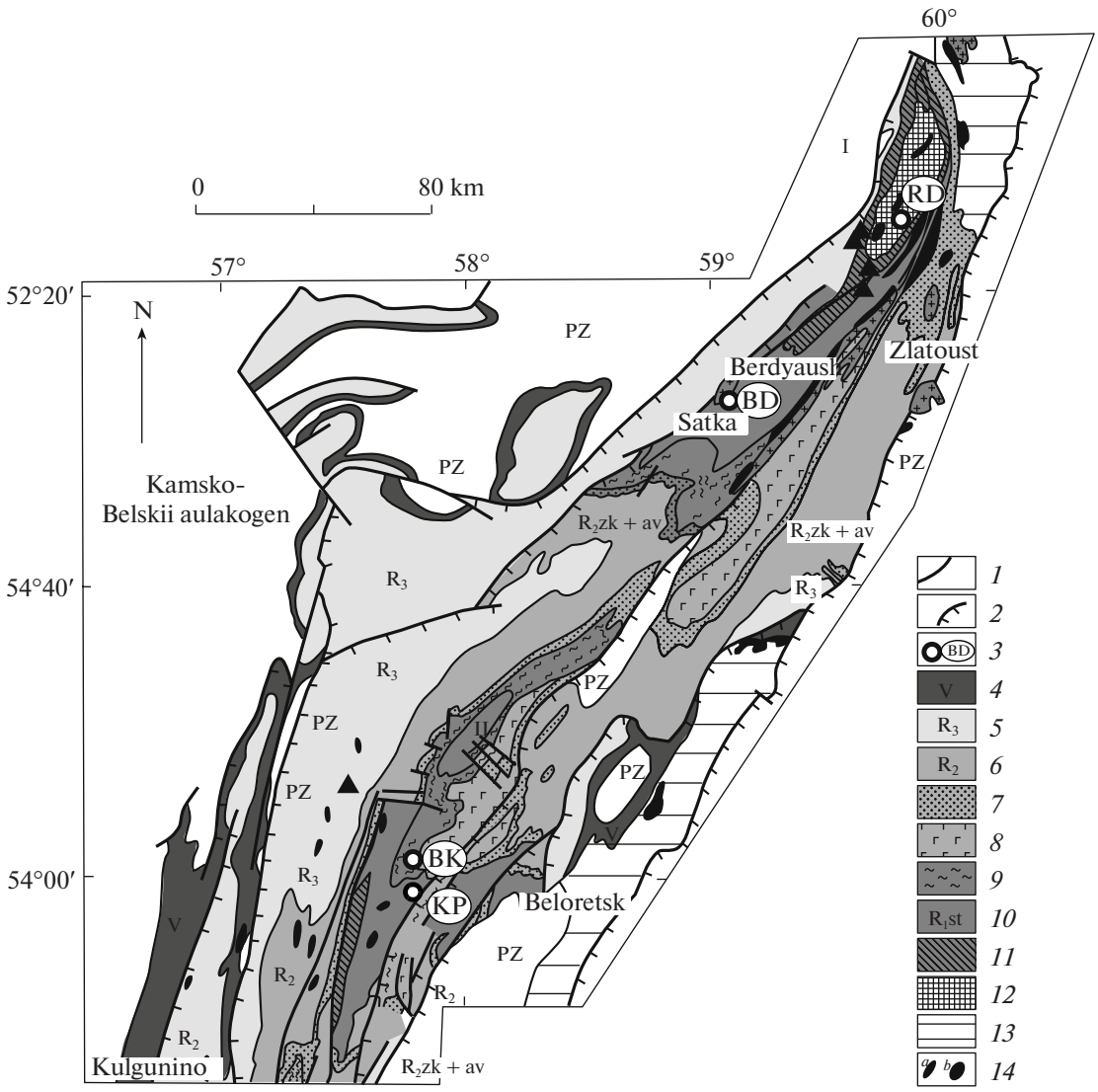


Fig. 2. Schematic geological map of the Bashkirian anticlinorium (Southern Ural) after [9] with localities of sampling: RD—Radostnyi Quarry, BD—Berdyaush Quarry, BK—Bolshoi Klyuch, KP—mount Kapkatash. *1*—Geological boundaries; *2*—main faults; *3*—localities of sampling; *4*—Vendian deposits; *5*—Upper Riphean deposits; *6*—Zigano-Komarovskaya suite (Middle Riphean); *7*—Zigalginskaya suite (Middle Riphean); *8*—Mashakskaya suite (Middle Riphean); *9*—Bakalskaya and Yushinskaya suites (Lower Riphean); *10*—Satkinskaya suite (Middle Riphean); *11*—Bolsheinzerskaya and Aiskaya suites (Lower Riphean); *12*—Taratash metamorphic complex; *13*—Ufaley and Uraltau metamorphic complexes; *14*—Magmatic complexes: gabbro-dolerites (*a*), granites (*b*).

suite rocks was determined as 1366 ± 6 Ma [10]; the Bakalskaya suite, 1385.3 ± 1.4 Ma [10, 11].

In basic rocks of North Priladozhie during detailed thermal and alternating field cleaning, carried out using the modern method [12], the high-temperature characteristic (following the origin of the coordinates) component of mainly north-northeast declination and negative inclination has been determined in the temperature range up to 580°C and alternating field of >40 mT. At the same time, in the central part of a thick dyke of the *B*-type, as well as in one of the Salma basalt flows, a component with reverse polarity has been established. These antipode components have an angle difference between directions $\gamma = 17.63°$ at the critical angle $\gamma_{cr} = 19.72°$ for Salma basalts and 10.72° at the critical angle $\gamma_{cr} = 13.43°$ for a dyke of the *B*-type.

The average direction of the high-temperature component established in magmatic rocks studied in North Priladozhie is close to that in granite-gneisses near the contact zone and is significantly different from the direction determined for rocks of the Ladozhskaya series, sampled 30 m from the chilled zone. The positive test for the rocks of the contact zone, as well as occurrence of the directions with reverse polarity in Salma basalts and the dyke of the *B*-type, speak for the originality of the determined high-temperature component of magnetization.

For the objects in the Southern Urals, the high-temperature component has been established in the

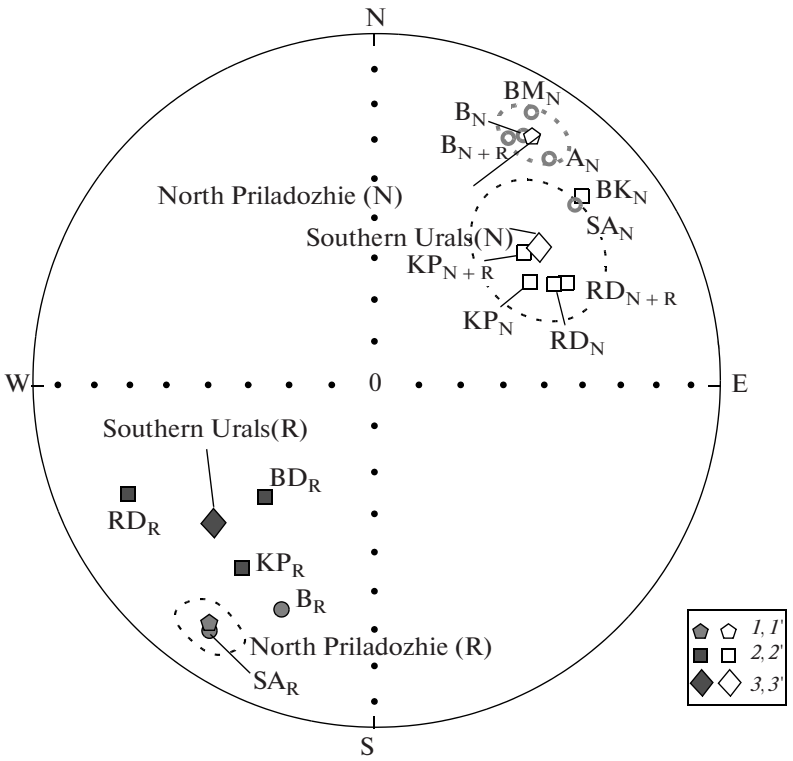


Fig. 3. Distribution of vectors of characteristic magnetization in the Mesoproterozoic magmatic rocks of the two areas studied. Directions of magnetization in rocks of North Priladozhie are shown by gray circles. *1, 1'*—average direction of the high-temperature component for four objects of North Priladozhie; *2, 2'*—directions of magnetization in rocks of the Southern Urals; *3, 3'*—average direction of the high-temperature component for four objects of the Southern Urals; (*1–3*) projection on lower hemisphere; (*1'–3'*) projection on upper hemisphere.

temperature interval of 400–580°C and alternating field interval of more than 40 mT. In magmatic rocks of the Radostny Quarry, Mount Kapkatash, and Bolshoi Klyuch Stream, the component is characterized by north-northeast declination and moderate negative inclination, whereas for the dyke in the Berdyaush Quarry, the component is characterized by southeast declination and moderate positive inclination. The positive test for inversion ($\gamma = 12.45°$ at a critical angle $\gamma_{cr} = 18.53°$) speaks for the originality of the determined component of magnetization.

The calculated average directions of magnetization of the high-temperature component for every object of these two localities are shown in Table 1 and Fig. 3.

The paleomagnetic poles calculated from the average direction of the high-temperature component in the Mesoproterozoic magmatic complexes of North Priladozhie and the Southern Urals are given in Table 2. Paleomagnetic poles established in the Mesoproterozoic magmatic complexes of North Priladozhie and the Southern Urals are close to those in

Table 2. Paleomagnetic poles for the East-European Craton

Object	Latitude of paleomagnetic pole	Longitude of paleomagnetic pole	*dp/dm* (α_{95})	Age, Ma	Data
	Angle degrees				
Khaukivesi dykes (KD)	**48**	**225**	**3**	**1840**	**[13]**
Ropruchei sill (RS)	**42**	**221**	**7**	**1770**	**[13]**
Aland dolerite dykes (AD)	**28**	**188**	**7**	**1577 ± 12**	**[1]**
Dalarna province (Tuna dykes ? TD)	11.2	183.1	3.4/6.3	1462 ± 5	[8]
Salma basalts (SA)	6	200	7	1499 ± 68	[5]
NORTH PRILADOZHIE (SP)	**15.2**	**177.1**	**7.3/3.8**	**1452 ± 12**	**Our data**
Valaamskii sill (VS)	13.8	166.4	1.7/3.4	1458 ± 4	[4]
Mean on dolerites (ID)	**4**	**158**	**4**	**1265**	**[1]**
SOUTHERN URALS (SU)	**1.8**	**193.0**	**19.2/11.4**	**1384 ± 2**	**Our data**

Note: *dp/dm* semiaxis of confidence oval, α_{95} radius of paleomagnetic pole confidence round. Key paleomagnetic poles are shown in bold.

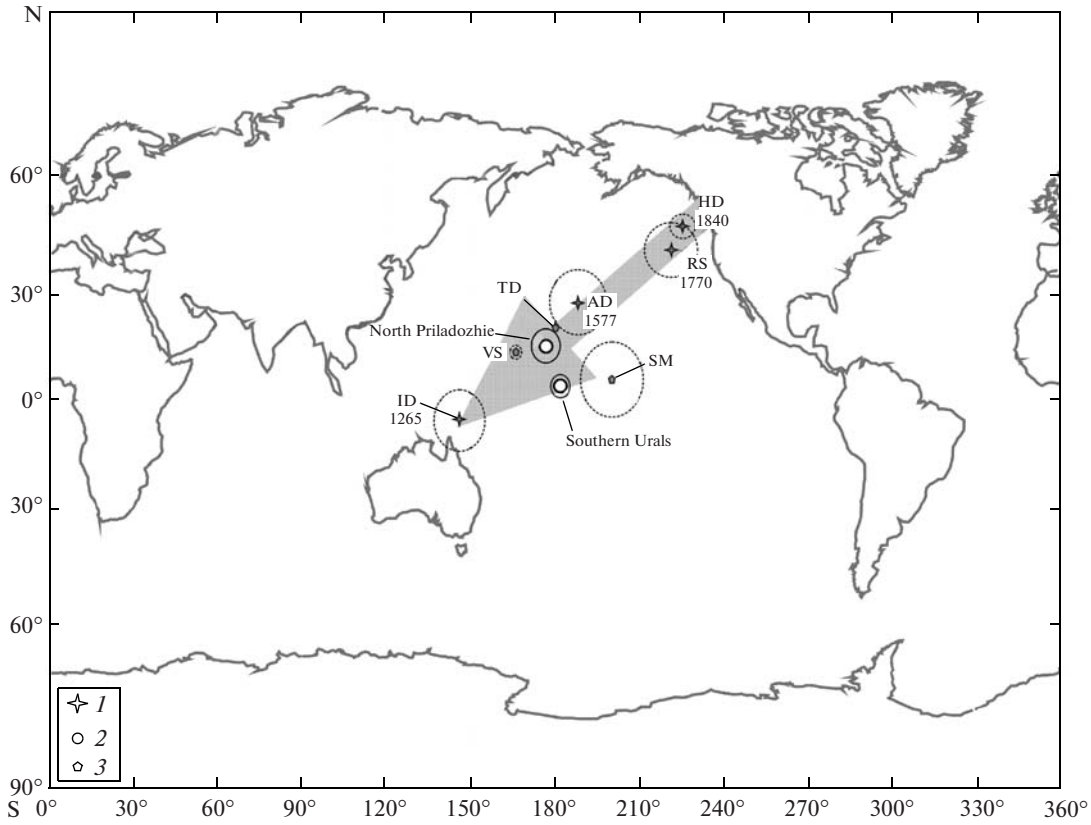


Fig. 4. Location of paleomagnetic poles of the East European Craton in the period of 1840–1265 Ma: *1*—key poles of the East European Craton after [1, 13]; *2*—paleomagnetic poles, obtained in this work; *3*—Mesoproterozoic poles [4, 5]. Letter symbols are given in Table 2.

magmatic complexes of the Dalarna Province (Central Sweden; Table 2) and lie on the trend of the key poles of East European Craton (Fig. 4). At the same time, they are different from the directions given in Table 2 in the Valaamskii sill [4] and Salma basalts [5]. It probably can be due to the fact that the secular variations in these works were not averaged.

Thus, as a result of paleomagnetic research carried out, new key Mesoproterozoic paleomagnetic poles in the East European Craton for two age intervals (1.45 and 1.38 Ga) have been determined. These intervals can be used for the further paleotectonic and paleogeographic reconstructions. According to the data obtained, the East European Craton in this interval was located in the near-equatorial latitudes in the Southern hemisphere and was subjected to counterclockwise rotation.

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